

Physics Equations Behind the Methane Simulator

Reference sheet (not part of the paper)

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The simulator is built on one core physical equation, the steady-state Gaussian plume, wrapped in standard atmospheric-dispersion and measurement models. Each scenario is a 500 m $\times$ 500 m site observed for 30 steps of 10 s each.

1 Gaussian plume dispersion

The concentration field is the closed-form solution of the steady-state advection–diffusion equation for a continuous point release, with total ground reflection handled by an image source:

$$C(x, y, z) = \frac{Q}{2\pi U \sigma_y(x) \sigma_z(x)} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left[\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right) \right] \quad (1)$$

where x is the downwind distance from the source, y the crosswind offset, z the receptor height, Q the release rate (kg/h), U the wind speed (floored at 0.5 m/s), $H = 3$ m the release height, and $z = z_r = 2$ m the sensor height. Concentration is zero upwind ($x \leq 0$). Since C is linear in Q , the simulator works with the unit-rate response

$$g = C/Q \quad (\text{ppm enhancement per kg/h}), \quad (2)$$

using 1 ppm $\text{CH}_4 = 6.55 \times 10^{-7}$ kg/m³ at 25°C.

2 Stability-dependent dispersion widths (Briggs / Pasquill–Gifford)

The plume half-widths grow with downwind distance according to the Briggs (1973) open-country curves, one pair per Pasquill–Gifford stability class A–F (A = very unstable/convective, F = very stable). Crosswind:

$$\sigma_y(x) = \frac{a_y x}{\sqrt{1 + b_y x}}, \quad a_y \in \{0.22, 0.16, 0.11, 0.08, 0.06, 0.04\}, \quad b_y = 10^{-4}. \quad (3)$$

Vertical, with a class-dependent functional form:

$$\sigma_z(x) = \begin{cases} c_z x & \text{classes A, B} \\ \frac{c_z x}{\sqrt{1 + d_z x}} & \text{classes C, D} \\ \frac{c_z x}{1 + d_z x} & \text{classes E, F} \end{cases} \quad (4)$$

Small source-size floors are added in quadrature so the plume has finite width at the source:

$$\sigma_y \leftarrow \sqrt{\sigma_y^2 + \sigma_{y0}^2}, \quad \sigma_z \leftarrow \sqrt{\sigma_z^2 + \sigma_{z0}^2}, \quad \sigma_{y0} = 2 \text{ m}, \quad \sigma_{z0} = 1 \text{ m}. \quad (5)$$

Each scenario draws one stability class uniformly from $\{A, \dots, F\}$.

3 Wind: AR(1) meander process

The true wind over the $T = 30$ steps is a mean vector modulated by a first-order autoregressive process on direction and log-speed. With mean speed $U_0 \sim \text{LogUniform}(1, 8)$ m/s and mean direction $\theta_0 \sim \text{Uniform}(0, 2\pi)$:

$$e_t^d = \rho e_{t-1}^d + \sqrt{1 - \rho^2} \varepsilon_t^d, \quad e_t^s = \rho e_{t-1}^s + \sqrt{1 - \rho^2} \varepsilon_t^s, \quad \varepsilon \sim \mathcal{N}(0, 1), \quad \rho = 0.7, \quad (6)$$

$$\theta_t = \theta_0 + A_{\text{dir}} e_t^d, \quad U_t = U_0 \exp(A_{\text{spd}} e_t^s), \quad (7)$$

with per-scenario meander amplitudes $A_{\text{dir}} \sim \text{Uniform}(20^\circ, 40^\circ)$ and $A_{\text{spd}} \sim \text{Uniform}(0.10, 0.30)$. The wind vector at step t is $\mathbf{u}_t = (U_t \cos \theta_t, U_t \sin \theta_t)$. This mimics short-term wind meander: it is what makes single-snapshot inversion ill-posed and time-resolved data informative.

The plume is *quasi-steady*: at each step the field is the steady-state solution for that step's wind (no puff transport lag, flat terrain, no obstacles).

4 Sensor observation model

Because the plume is linear in Q , each reading is

$$y_{s,t} = Q g_{s,t}(\mathbf{x}_{\text{src}}, \mathbf{u}_t, \text{stab}) + \varepsilon_{s,t}, \quad \varepsilon_{s,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (8)$$

with per-scenario draws

$$Q \sim \text{LogUniform}(10, 500) \text{ kg/h}, \quad \sigma \sim \text{LogUniform}(0.2, 3.0) \text{ ppm}, \quad n_{\text{sensors}} \sim \text{Uniform}\{4, \dots, 12\}, \quad (9)$$

sensor positions uniform on the site, and independent per-reading dropout with probability $p_d \sim \text{Uniform}(0, 0.2)$ (missing data). The noise range spans low-cost metal-oxide to mid-grade sensors; readings are ppm enhancement above background.

5 Anemometer error model (what the method actually sees)

The localizer never receives the true wind. The measured wind is the truth corrupted per time step by Gaussian direction error and lognormal speed error:

$$\hat{\theta}_t = \theta_t + \delta_t, \quad \delta_t \sim \mathcal{N}(0, \sigma_\theta^2), \quad \hat{U}_t = U_t e^{\eta_t}, \quad \eta_t \sim \mathcal{N}(0, \sigma_s^2), \quad (10)$$

with $\sigma_\theta = 10^\circ$ and $\sigma_s = 0.10$ (10% speed error). The physics input maps are computed under $K = 8$ independent draws from this same error model applied to the measured wind, which is what the fragility and fit-quality channels respond to.

Summary

Scenarios are generated from a steady-state Gaussian plume with ground reflection and Briggs–Pasquill–Gifford stability-dependent dispersion, driven by an autoregressive meandering wind, observed through a linear sensor model with Gaussian noise and dropout; the localizer only ever sees an anemometer-corrupted version of that wind.

Equation	Code location	Role
(1)–(2) plume	<code>src/methane_model.py:53</code>	forward model g
(3)–(5) Briggs σ 's	<code>src/methane_model.py:29--50</code>	dispersion widths
(6)–(7) AR(1) wind	<code>src/methane_t.py:40--57</code>	true wind sequence
(8)–(9) observations	<code>src/methane_t.py:111--116</code>	readings y
(10) anemometer error	<code>src/methane_t_uncertain.py:62--66</code>	measured wind